

Contribution Record — Smart Scientific Calculator

Repository: <https://github.com/Krishnarajss/Smart-Scientific-Calculator> (<https://github.com/Krishnarajss/Smart-Scientific-Calculator>)

1. Team Members

#	Name	Registration No.	GitHub Username	Role / Responsibility
1	Siddhartha Tiwari	20243275	princetiwari1204	Team Leader — Full-stack implementation (all iterations)
2	Krishnaraj Singh	20243278	Krishnarajss	SRS, Architecture, UML Diagrams, Use Cases, Repo Setup
3	Sourabh Verma	20243279	—	Testing (Test Cases & Evidence), Project Management Docs
4	Umar Rashid	20243591	—	Demo Video, Mid-Semester Project Report

2. Individual Contribution Summary

Siddhartha Tiwari (20243275) — Team Leader

- Implemented the **complete source code** across all six iterations:
 - Iteration 1:** Basic arithmetic operations (+, -, *, /, %, AC, DEL)
 - Iteration 2 & 3:** Scientific functions ($\sqrt{}$, sin, cos, x^y)
 - Iteration 4:** Error handling (division by zero, invalid domain, syntax errors)
 - Iteration 5:** Responsive grid-based UI layout, real-time display, button feedback
 - Iteration 6:** Node.js/Express backend server, MongoDB integration via Mongoose, history API
- Authored frontend files: `public/index.html`, `public/script.js`, `public/style.css`
- Authored backend: `server.js`
- Configured project dependencies: `package.json`, `package-lock.json`
- Authored `Readme.md` (project documentation)
- Created and merged PR #1 adding the full project structure

Krishnaraj Singh (20243278)

- Created and owns the GitHub repository (`Krishnarajss/Smart-Scientific-Calculator`)

- Added the initial project description to `Readme.md`
- Prepared and uploaded all requirements and design artifacts:
 - `SRS_document.pdf` — Software Requirements Specification
 - `Architecture_Diagram_Clean.pdf` — System architecture diagram
 - `UML_Diagrams.pdf` — UML class/sequence diagrams
 - `Use_Cases_or_User_Stories.pdf` — Use-case descriptions and user stories
- Maintained the `SRS` branch with all documentation files
- Uploaded `server.js` to the `SRS` branch

Sourabh Verma (20243279)

- Designed and documented **19 test cases** covering all six iterations:
 - Basic arithmetic (TC-001 to TC-006)
 - Scientific functions (TC-007 to TC-010)
 - Error handling (TC-011 to TC-013)
 - UI and interaction (TC-014 to TC-015)
 - Database history integration (TC-016 to TC-019)
- Created the `Test_Cases.md` document with detailed steps, expected/actual results, and pass/fail status
- Created the `Test_Evidence.md` document with screenshot placeholders and evidence structure
- Prepared all **Project Management** documents:
 - `Project_Plan.md` — Timeline, milestones, task allocation, risks, and remaining work
 - `Meeting_Log.md` — Six meeting records with agendas, decisions, and action items
 - `Contribution_Record.md` — This document

Umar Rashid (20243591)

- Recorded and produced the **demonstration video** (5–7 minutes):
 - Project title and team introduction
 - Problem explanation and proposed solution overview
 - Architecture walkthrough
 - Live demo of working features
 - Current limitations and remaining work
 - Individual contribution statements
- Uploaded the video link to `07_Demo/DEMO_VIDEO_LINK.txt`
- Authored the **Mid-Semester Project Report** (`01_Report/MidSem_Project_Report.pdf`):
 - Cover page, problem statement, objectives, scope
 - Requirements summary, system design overview
 - Implementation progress with screenshots
 - Testing summary, project management status
 - Challenges, future work, and references

3. Contribution Percentage

Member	Reg. No.	Contribution (%)
Siddhartha Tiwari	20243275	25%
Krishnaraj Singh	20243278	25%
Sourabh Verma	20243279	25%
Umar Rashid	20243591	25%
Total		100%

4. Git Commit History

The following is the actual commit history extracted from the GitHub repository.

main Branch

#	SHA	Author	Date	Message
1	6a071bf	Krishnaraj Singh	2 Oct 2026, 12:57 PM	Add initial project description to Readme.md
2	27cbe28	Siddhartha Tiwari	2 Oct 2026, 5:55 PM	Add full project structure across all iterations
3	cff6fdc	Siddhartha Tiwari	2 Oct 2026, 6:16 PM	Merge pull request #1 from Krishnarajss/princetiware

SRS Branch

#	SHA	Author	Date	Message
4	8d0ea6c	Krishnaraj Singh	2 Oct 2026, ~3:30 PM	Add files via upload (Architecture_Diagram_Clean.pdf)
5	2e9a152	Krishnaraj Singh	2 Oct 2026, ~3:30 PM	Add files via upload (SRS document.pdf)
6	1394bd2	Krishnaraj Singh	2 Oct 2026, ~3:30 PM	Add files via upload (Use_Cases_or_User_Stories.pdf)
7	8da4bfa	Krishnaraj Singh	2 Oct 2026, ~3:00 PM	Add files via upload (UML_Diagrams.pdf, server.js)

Contributor Statistics (from GitHub)

Contributor	GitHub Username	Commits	Files Changed
Siddhartha Tiwari	princetiware1204	2	public/, node_modules/, Readme.md, package.json, package-lock.json
Krishnaraj Singh	Krishnarajss	5	Readme.md, SRS document.pdf, Architecture_Diagram_Clean.pdf, UML_Diagrams.pdf, Use_Cases_or_User_Stories.pdf, server.js
Sourabh Verma	—	—	05_Testing/, 06_Project_Management/ (local, pending push)

Contributor	GitHub Username	Commits	Files Changed
Umar Rashid	—	—	01_Report/, 07_Demo/ (local, pending push)

5. Task Allocation Table

Task	Assigned To	Status
Repository creation and setup	Krishnaraj Singh	☐ Completed
SRS document	Krishnaraj Singh	☐ Completed
Use-case diagram / User stories	Krishnaraj Singh	☐ Completed
Architecture diagram	Krishnaraj Singh	☐ Completed
UML diagrams	Krishnaraj Singh	☐ Completed
Iteration 1: Basic arithmetic implementation	Siddhartha Tiwari	☐ Completed
Iteration 2 & 3: Scientific functions	Siddhartha Tiwari	☐ Completed
Iteration 4: Error handling	Siddhartha Tiwari	☐ Completed
Iteration 5: UI design and interaction	Siddhartha Tiwari	☐ Completed
Iteration 6: Backend server + MongoDB	Siddhartha Tiwari	☐ Completed
Test-case design and execution (19 cases)	Sourabh Verma	☐ Completed
Test evidence documentation	Sourabh Verma	☐ Completed
Project plan	Sourabh Verma	☐ Completed
Meeting log	Sourabh Verma	☐ Completed
Contribution record	Sourabh Verma	☐ Completed
Mid-semester project report	Umar Rashid	☐ Completed
Demonstration video	Umar Rashid	☐ Completed
Final ZIP packaging and submission	All members	☐ In Progress

6. Weekly Progress Record

Week	Tasks Completed	Members Involved
Week 1–2	Topic selection, problem statement, initial research, SRS drafting	All members
Week 3	SRS document finalised, use-case diagram, architecture design	Krishnaraj Singh
Week 4	Iterations 1–3 implemented (basic arithmetic + scientific functions)	Siddhartha Tiwari
Week 5	Iteration 4 (error handling) implemented	Siddhartha Tiwari
Week 6	Iteration 5 (responsive UI) implemented	Siddhartha Tiwari
Week 7	Iteration 6 (Express + MongoDB) implemented	Siddhartha Tiwari
Week 8	UML diagrams prepared, architecture diagram finalised	Krishnaraj Singh
Week 9	Testing (19 test cases), project management docs, report, demo video	Sourabh Verma, Umar Rashid

7. Individual Contribution Declaration

By listing our names below, each team member confirms that the contributions described in this document are accurate and that all members participated meaningfully in the project.

Member	Reg. No.	Signature / Confirmation Date
Siddhartha Tiwari	20243275	
Krishnaraj Singh	20243278	
Sourabh Verma	20243279	
Umar Rashid	20243591	

Meeting Log — Smart Scientific Calculator

Instructions: Fill in the attendees, discussion points, and action items for each meeting. Add new entries as meetings occur.

Meeting 1

Field	Detail
Date	[Fill in: e.g. 12 Aug 2026]
Time	[Fill in: e.g. 2:00 PM – 3:00 PM]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Initial brainstorming and project-topic selection.
2. Discussion of feasible technology stacks.
3. Assignment of preliminary research tasks.

Discussion Summary

- The team decided to build a **Smart Scientific Calculator** as the project topic.
- Agreed on **HTML/CSS/JS** for the frontend and **Node.js + Express + MongoDB** for the backend.
- Discussed the incremental development model as the process framework.

Decisions Made

- Project topic finalised: Smart Scientific Calculator.
- Development model: Incremental.
- Technology stack: HTML, CSS, JS, Node.js, Express, MongoDB.

Action Items

Action	Owner	Deadline
Draft initial problem statement	[Member 1]	[Date]
Research similar calculator applications	[Member 2]	[Date]
Set up project repository	[Member 3]	[Date]
Create initial UI wireframe sketch	[Member 4]	[Date]

Meeting 2

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Review problem statement and objectives.
2. Define functional and non-functional requirements.
3. Plan Iteration 1 scope and task distribution.

Discussion Summary

- Reviewed and refined the problem statement.
- Identified six user needs (UN-01 to UN-06) to be addressed across six iterations.
- Finalised functional requirements: basic arithmetic, scientific functions, error handling, responsive UI, and database history.
- Finalised non-functional requirements: performance (< 1s response), usability, browser compatibility.

Decisions Made

- SRS document structure agreed upon.
- Iteration 1 scope: basic arithmetic operations (+, -, *, /, %, AC, DEL).

Action Items

Action	Owner	Deadline
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Action	Owner	Deadline
Write SRS document	[Member 1], [Member 2]	[Date]
Design use-case diagram	[Member 2]	[Date]
Implement Iteration 1 (basic arithmetic)	[Member 1], [Member 3]	[Date]
Design CSS layout for calculator grid	[Member 4]	[Date]

Meeting 3

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Demo of Iteration 1 (basic arithmetic).
2. Plan Iterations 2 & 3 (scientific functions).
3. Review architecture diagram.

Discussion Summary

- Iteration 1 was demonstrated successfully — all basic operations work.
- Discussed scientific function integration: `sqrt`, `sin`, `cos`, `x^y`.
- Reviewed the proposed 3-tier architecture (Frontend → Express Server → MongoDB).

Decisions Made

- Iteration 1 accepted.
- Scientific functions will use degree-based input (not radians).
- Architecture diagram approved.

Action Items

Action	Owner	Deadline
Implement <code>sqrt</code> , <code>sin</code> , <code>cos</code>	[Member 2]	[Date]
Implement <code>x^y</code> (power function)	[Member 4]	[Date]
Finalise architecture diagram for report	[Member 3]	[Date]

Meeting 4

Field	Detail
Date	[Fill in]

Field	Detail
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Demo of Iterations 2 & 3 (scientific functions).
2. Plan Iteration 4 (error handling).
3. Discuss UI improvements for Iteration 5.

Discussion Summary

- Scientific functions demonstrated and working.
- Identified edge cases: division by zero, negative square root, malformed expressions.
- Discussed responsive grid layout and button feedback for better UX.

Decisions Made

- Iteration 4 to use try-catch blocks with user-friendly "Error" messages.
- Iteration 5 to use CSS Grid for the keypad and real-time display updates.

Action Items

Action	Owner	Deadline
Implement error handling (Iteration 4)	[Member 1]	[Date]
Redesign UI with CSS Grid (Iteration 5)	[Member 4]	[Date]
Begin database schema design	[Member 3]	[Date]

Meeting 5

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Demo of Iterations 4 & 5.
2. Plan Iteration 6 (history & database).
3. Plan test-case design.

Discussion Summary

- Error handling and UI improvements demonstrated successfully.
- Discussed MongoDB schema: { `expression`, `result`, `createdAt` }.
- Planned REST API endpoints: `POST /api/history`, `GET /api/history`.

Decisions Made

- Use Mongoose for schema validation and MongoDB interaction.
- History retrieval limited to 10 most recent records, sorted by `createdAt` descending.

Action Items

Action	Owner	Deadline
Implement Express server + MongoDB (Iter. 6)	[Member 3]	[Date]
Connect frontend to API	[Member 1]	[Date]
Design test cases for all iterations	[Member 2], [Member 4]	[Date]

Meeting 6

Field	Detail
Date	[Fill in]
Time	[Fill in]
Mode	[Online / In-person]
Attendees	[Member 1], [Member 2], [Member 3], [Member 4]

Agenda

1. Full integration demo (all 6 iterations).
2. Review test results.
3. Plan mid-semester submission packaging.

Discussion Summary

- Full end-to-end demo completed: basic arithmetic → scientific → error handling → UI → database history.
- All 19 test cases passed.
- Discussed mid-semester submission structure and report writing assignments.

Decisions Made

- Each member assigned specific report sections.
- Demonstration video to be recorded before submission deadline.

Action Items

Action	Owner	Deadline
Write report sections (assigned)	All members	1 Oct 2026
Record demonstration video	[Member 1], [Member 3]	2 Oct 2026
Package ZIP for submission	[Member 2]	3 Oct 2026
Final review and submission	All members	3 Oct 2026

Reminder: Replace all [Fill in] and [Member N] placeholders with actual dates and names before submission.

Project Plan — Smart Scientific Calculator

1. Project Overview

Field	Detail
Project Title	Smart Scientific Calculator
Development Model	Incremental Software Development Model
Technology Stack	HTML, CSS, JavaScript (Frontend) · Node.js, Express (Backend) · MongoDB via Mongoose (Database)
Team Size	[Fill in: e.g. 4 members]
Start Date	[Fill in: e.g. August 2026]
Mid-Sem Deadline	3 October 2026
Final Deadline	[Fill in]

2. Development Iterations & Timeline

The project follows six incremental iterations. Each iteration adds a well-defined set of features on top of the previous increment.

Iteration	Feature Area	Planned Start	Planned End	Status
1	Basic Arithmetic (UN-01): +, -, *, /, %, AC, DEL	Week 1	Week 2	☐ Completed

Iteration	Feature Area	Planned Start	Planned End	Status
2 & 3	Scientific Functions (UN-02 & UN-03): $\sqrt{}$, $\sin$, $\cos$, x^y	Week 3	Week 4	☐ Completed
4	Error Handling (UN-04): Division by zero, invalid domain, syntax errors	Week 5	Week 5	☐ Completed
5	UI & Interaction (UN-05): Responsive grid layout, real-time display, button feedback	Week 6	Week 6	☐ Completed
6	History & Database (UN-06): Express server, MongoDB CRUD, history view	Week 7	Week 8	☐ Completed
—	Testing, Documentation & Final Polish	Week 9	Week 12	☐ In Progress

3. Milestone Tracker

Milestone	Target Date	Status
Problem statement finalised	Week 1	☐ Done
Requirements (SRS) document ready	Week 2	☐ Done
Architecture & DB design approved	Week 3	☐ Done
Iteration 1 demo-ready	End of Week 2	☐ Done
Iteration 2 & 3 demo-ready	End of Week 4	☐ Done
Iteration 4 demo-ready	End of Week 5	☐ Done
Iteration 5 demo-ready	End of Week 6	☐ Done
Iteration 6 demo-ready	End of Week 8	☐ Done
Mid-semester submission	3 Oct 2026	☐ In Progress
Final testing & polish	Week 10	☐ Planned
Final submission	[Fill in]	☐ Planned

4. Task Allocation Table

Task	Assigned To	Status
Requirements gathering & SRS	[Member 1], [Member 2]	☐ Done
Use-case diagram / User stories	[Member 2]	☐ Done
Architecture & database design	[Member 3]	☐ Done
UI wireframe & prototype	[Member 4]	☐ Done
Iteration 1 implementation	[Member 1], [Member 3]	☐ Done
Iteration 2 & 3 implementation	[Member 2], [Member 4]	☐ Done
Iteration 4 implementation	[Member 1]	☐ Done
Iteration 5 implementation	[Member 4]	☐ Done
Iteration 6 implementation	[Member 3]	☐ Done
Test-case design & execution	[Member 2], [Member 4]	☐ Done
Mid-semester report writing	All members	☐ In Progress
Demonstration video	[Member 1], [Member 3]	☐ Planned

Note: Replace [Member N] with actual team-member names.

5. Risks and Mitigation

#	Risk	Likelihood	Impact	Mitigation Strategy
1	MongoDB unavailable on deployment machine	Medium	High	Provide Docker Compose file; document manual setup steps
2	<code>eval()</code> in front-end poses security risk	Medium	Medium	Planned: replace with a safe expression parser in final phase
3	Floating-point precision in scientific functions	Low	Low	Document known tolerance; round display to 10 decimal places
4	Team-member unavailability near deadline	Low	High	Maintain shared repo; any member can pick up pending work
5	Moodle upload size limit exceeded	Low	Medium	Host demo video externally; include <code>DEMO_VIDEO_LINK.txt</code>

6. Delayed or Pending Tasks

Task	Original Target	Revised Target	Reason for Delay
Expression parser to replace <code>eval()</code>	Week 8	Post mid-sem	Prioritised DB integration
Keyboard input support	Week 7	Post mid-sem	Scope deferred
Unit tests with Jest (automated)	Week 9	Post mid-sem	Manual testing done first

7. Work Planned for Remaining Semester

1. Replace `eval()` with a secure mathematical expression parser.
2. Add keyboard input support for desktop users.
3. Implement automated unit tests using **Jest**.
4. Add a proper in-page history panel (instead of browser `alert()`).
5. Deploy the application to a cloud platform (e.g. Render / Railway).
6. Conduct usability testing with peer users.
7. Final documentation and submission packaging.

